

Li Zhoujian

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[Personal Homepage Link: <https://haydenil.github.io/Resume/>

Education

National University of Singapore 2025.08-Present

- **Master** of Science in Robotics
- Key Course:Robot Vision and AI, Robot Kinematics, Robot Dynamics and Control, Machine Learning

Soochow University 2020.09-2024.09

- **Bachelor** of Science in Intelligent Manufacturing Engineering
- Key Course:Robotics, Artificial Intelligence, Java Programming, PLC & Electric Control

Research Projects

Below are the primary research projects. A comprehensive list can be found on my personal website :)

Multi-Goal Mobile Robot Mapping and Navigation System 2026.01-2026.03

- ◆ **Overview:** Built a full mapping-and-navigation stack in ROS/Gazebo for multi-goal missions in dynamic indoor environments, covering 3D perception, mapping, localization, and planning.
- **Challenges & Solutions:**1) Built 3D point-cloud maps with A-LOAM and generated 2D occupancy grids for navigation. 2) Fused AMCL with an EKF to suppress long-term localization drift; added YOLO for real-time target detection. 3) Tuned A*/TEB global – local planning and multi-layer costmaps, achieving a 100% success rate in autonomous goal search and navigation in a virtual factory.

RL-Based Single-Leg Balancing for Quadruped Robots 2025.08-2025.11

- ◆ **Overview:** Developed a PPO-based single-leg balance controller achieving handstand and single-contact balancing in highly underactuated extreme postures where model-based control lacks robustness.
- **Challenges & Solutions:**1) Redesigned rewards around CoM alignment and strict contact topology to overcome sparse rewards and local optima. 2) Stabilized training with staged curricula, GAE, and a KL-adaptive learning rate in a custom PPO framework. 3) Validated robustness via Isaac Lab→MuJoCo.

Internship Experience

Galbot Humanoid Locomotion Algorithm Intern 2026.05-Present

- ◆ **Responsibilities:** RL-based locomotion control development and real-robot deployment for the first-generation humanoid NX1, covering simulation training, perception & mapping, onboard inference, and real-robot control.
- **Key Contributions:**1) Trained perceptive policies in MuJoCo with an asymmetric actor-critic fusing proprioception and height-scan observations. 2) Learned natural walking and stair-climbing gaits from AMP motion priors on GMR-retargeted motion capture. 3) Narrowed the sim2real gap via terrain curricula, domain randomization, and height-map noise injection. 4) Deployed onboard: LiDAR – inertial odometry on Jetson AGX feeds real-time elevation maps to the ONNX policy in a 30 Hz control loop. 5) NX1 walks at 0.8 m/s with strong push recovery and climbs stairs continuously.

Publications & Awards

DVDP: An End-to-End Policy for Mobile Robot Visual Docking with RGB-D Perception	Co-first author
TG-RRT*: Enhanced Learning-Based Optimal Path Planning via Transformer-CNN Hybrid Network and Goal-Directed Strategy	Co-first author
A Reinforcement Learning-Driven Algorithm for Rapid Path Replanning of Robot Navigation in Indoor Uncertain Discrete Environments	CCDC 2025
RAICOM Robot Developer Competition 2023, National Finals	National First Prize
The 24th China Robot and Artificial Intelligence Competition, National Finals	National Second Prize

Additional information

- **Programming:** Python, C++, Linux/Ubuntu, ROS; **physics simulators:** Gazebo, Isaac Lab, MuJoCo.
- RL locomotion control for humanoids/quadrupeds: PPO optimization, reward & curriculum design for underactuated systems, RSL-RL, sim2sim validation.
- Long-term robotics/UAV research experience; independently reproduces state-of-the-art methods from English literature.